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12. *(a)* A student planned to distinguish between the following eight compounds.

magnesium hydroxide	magnesium carbonate
iron(II) hydroxide	iron(II) carbonate
chromium(III) hydroxide	chromium(III) carbonate
lead(II) hydroxide	lead(II) carbonate

He used the following method.

Step 1 Add dilute acid until all the solid has disappeared. Record any effervescence.

Step 2 Add 1 cm^3 of sodium hydroxide solution to each solution formed in step 1. Record any precipitate observed.

(i) State which compound(s) would give effervescence and why they do so. [2]

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(ii) The student plans to use dilute hydrochloric acid in step 1. His teacher tells him that this is **not** the correct acid to use.

Explain why hydrochloric acid should not be used and suggest an appropriate acid to use in its place. [2]

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(iii) Give the colours of the precipitates formed when sodium hydroxide solution is added to solutions containing the following ions. [2]

$Mg^{2+}(aq)$ ....

$Fe^{2+}(aq)$ ....

$Cr^{3+}(aq)$ ....

$Pb^{2+}(aq)$ ....

(iv) The method given opposite is incomplete.

Suggest an additional step that would allow the remaining solutions formed in step 2 to be identified. Give the expected observations. State the property of the metals that allows them to be distinguished in this way. [3]

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(b) Many rocks that contain carbonate anions also contain a mixture of cations. Atomic absorption spectroscopy can be used to find the ratio of the amounts of different cations present.

Analysis of an acid solution formed from the carbonate mineral huntite shows that it contains 91 $ng\,cm^{-3}$ of magnesium ions and 50 $ng\,cm^{-3}$ of calcium ions. These are the only two metal cations present.

(i) Find the formula of the mineral huntite. [3]

Formula ....

(ii) The initial solution was prepared using 220 μg of huntite. Calculate the volume of aqueous acid used to form the solution for analysis. [3]

Volume = ... dm^3

15

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